→ learn

- Network architecture
- Network operation
- Network attack

Module - 1

structure of network
standard networking tool
Cloud network
TCP/IP Model

What are network?

A group of connected devices

LAN → school, Home
WAN → Country State

Network tool

HUB:- A network device broadcast info to every device on the network

Switch :- a device connection between specific devices on network by sending and receiving data

Router :- A Network device that connect multiple networks together

Modem :- device connect my router to the internet and access to lan

Firewall :- A network security device that monitor traffic to or from your network first line of defense.

WAP (wireless access point) :- A WAP send signal over radio waves creating wireless network

Cloud network => collection of server

Cloud computing is practice of using remote serve
application, and network service that are
hosted on internet instead on local devices

network Communication

DataPacket : A basic unit of information
that travels frome one device
to another within a network

formate of a Packet

IP address /
mac adress /
Protocol number

Bandwidth

the amount of data a device receive every second

Speed :- the rate at with data packet are received or downloaded

Packet Sniffing :- the practice of capturing and inspecting data packets across a network

TCP/IP model

↳ Internet protocol

transmition Control protocol

Port : a software based location that organize the sending and receiving data between devices on network

Port Number :- assign priority

Port 25 - Email

Port 443 - secure internet comm

Port 20 - large file transfer

4 layer of TCP/IP model

TCP/IP model is a framework used to visulize how data is organized and transmitted across the network

1. Network access layer
2. Internet layer
3. Transport Layer
4. Application Layer

Osi model

OSI model is a standardized concept that describe seven layer of communication

Application Layer ::_7

The application layer include processes that directly involve everyday user .include all network protocol that software application use to connect to internet

Presentation layer ::— 6

Function at presentation layer is involve data translation and encryption for network
]]]

Session Layer :5
DESCRIBES WHEN THE CONNECTION BETWEEN TWO DEVICE IS ESATBLISHED

~~TRANSPORTA~~

Transport level :. 4

responsible for delivering data between devic

Network Layer . . 3

oversees receiving the frames frome the data link layer and delivers them to intended destination

Data link Layer . . 2

organise sending and recivinig data within single network

Physical layer

Physical hardware involved in network transmission

IP address

A unique string of character that identifies the location of device on the internet

two types

1. IP version 4 (IPV4)

2. IP version 6 (IPV6)

MAC address

A unique alpha numeric identifier that is assigned to each physical device on a network

Terms and definitions from Course 3, Module 1

Bandwidth: The maximum data transmission capacity over a network, measured by bits per second

Cloud computing: The practice of using remote servers, application, and network services that are hosted on the internet instead of on local physical devices

Cloud network: A collection of servers or computers that stores resources and data in remote data centers that can be accessed via the internet

Data packet: A basic unit of information that travels from one device to another within a network

Hub: A network device that broadcasts information to every device on the network

Internet Protocol (IP): A set of standards used for routing and addressing data packets as they travel between devices on a network

Internet Protocol (IP) address: A unique string of characters that identifies the location of a device on the internet

Local Area Network (LAN): A network that spans small areas like an office building, a school, or a home

Media Access Control (MAC) address: A unique alphanumeric identifier that is assigned to each physical device on a network

Modem: A device that connects your router to the internet and brings internet access to the LAN

Network: A group of connected devices

Open systems interconnection (OSI) model: A standardized concept that describes the seven layers computers use to communicate and send data over the network

Packet sniffing: The practice of capturing and inspecting data packets across a network

Port: A software-based location that organizes the sending and receiving of data between devices on a network

Port: A software-based location that organizes the sending and receiving of data between devices on a network

Router: A network device that connects multiple networks together

Speed: The rate at which a device sends and receives data, measured by bits per second

Switch: A device that makes connections between specific devices on a network by sending and receiving data between them

TCP/IP model: A framework used to visualize how data is organized and transmitted across a network

Transmission Control Protocol (TCP): An internet communication protocol that allows two devices to form a connection and stream data

User Datagram Protocol (UDP): A connectionless protocol that does not establish a connection between devices before transmissions

Wide Area Network (WAN): A network that spans a large geographic area like a city, state, or country

Network operation

Network protocol :- a set of rule used by two or more device on a network to describe the order of delivery and structure of data

Address Resolution Protocol (ARP)

A network protocol used to determine the MAC address of the next router or device on the path

Domain name System (DNS)

A network protocol that translate internet domain names into IP address

Management Protocol

Simple Network management Protocol (SNMP) used for managing device on network. It can reset Password on the network or change it to its baseline Configuration

ICMP Internet control message protocol
used by device to tell each other about data transmission error accross the network

Security layer

Hyper text transfer protocol secure (HTTPS)
Secure version of HTTP that use secure Socket layer / transport layer

Secure file transfer protocol (SFTP)
used to transfer file over network
Use secure shell (SSH) tough tcp port - 22

Protocol	Port
DHCP	UDP port 67 (servers) UDP port 68 (clients)
ARP	none
Telnet	TCP port 23
SSH	TCP port 22
POP3	TCP/UDP port 110 (unencrypted) TCP/UDP port 995 (encrypted, SSL/TLS)
IMAP	TCP port 143 (unencrypted) TCP port 993 (encrypted, SSL/TLS)
SMTP	TCP/UDP Port 25 (unencrypted)
SMTPS	TCP/UDP port 587 (encrypted, TLS)

Wireless Protocol

IEEE 802.11 (WiFi)

A set of communication standards for wireless lan

WPA (wifi protected access)

A wiless security protocol for device to connect to internet

WPA2 & WPA3

WPA2

The second version of Wi-Fi Protected Access—known as WPA2—was released in 2004. WPA2 improves upon WPA by using the Advanced Encryption Standard (AES). WPA2 also improves upon WPA's use of TKIP. WPA2 uses the Counter Mode Cipher Block Chain Message Authentication Code Protocol (CCMP), which provides encapsulation and ensures message authentication and integrity. Because of the strength of WPA2, it is considered the security standard for all Wi-Fi transmissions today. WPA2, like its predecessor, is vulnerable to KRACK attacks. This led to the development of WPA3 in 2018.

Personal

WPA2 personal mode is best suited for home networks for a variety of reasons. It is easy to implement, initial setup takes less time for personal than enterprise version. The global passphrase for WPA2 personal version needs to be applied to each individual computer and access point in a network. This makes it ideal for home networks, but unmanageable for organizations.

Enterprise

WPA2 enterprise mode works best for business applications. It provides the necessary security for wireless networks in business settings. The initial setup is more complicated than WPA2 personal mode, but enterprise mode offers individualized and centralized control over the Wi-Fi access to a business network. This means that network administrators can grant or remove user access to a network at any time. Users never have access to encryption keys, this prevents potential attackers from recovering network keys on individual computers.

WPA3

WPA3 is a secure Wi-Fi protocol and is growing in usage as more WPA3 compatible devices are released. These are the key differences between WPA2 and WPA3:

- WPA3 addresses the authentication handshake vulnerability to KRACK attacks, which is present in WPA2.
- WPA3 uses Simultaneous Authentication of Equals (SAE), a password-authenticated, cipher-key-sharing agreement. This prevents attackers from downloading data from wireless network connections to their systems to attempt to decode it.
- WPA3 has increased encryption to make passwords more secure by using 128-bit encryption, with WPA3-Enterprise mode offering optional 192-bit encryption.

Firewall

↳ Can use

- Port filtering → A function that blocks or allow certain port number to limit unwanted communication

How firewall operates?

stateful :- A class of firewall that keeps track of information passing through it and proactively filters out threat

stateless :- operate based on predefined rules and does not keep track of information from data packets

VPN

virtual private network

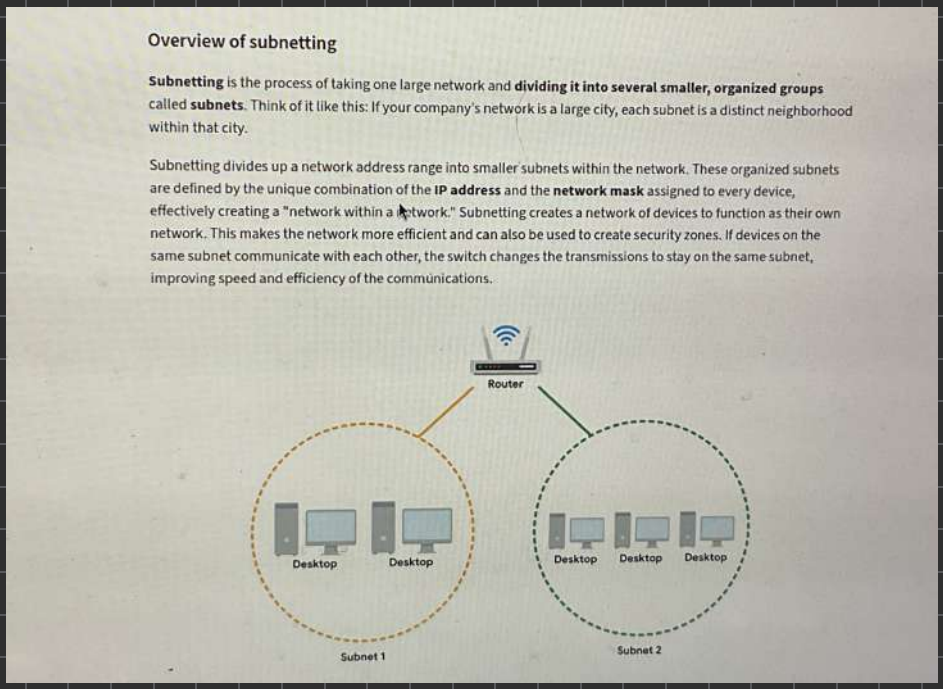
Network security service that change my public IP and hide my virtual location to keep data private

VPN Encapsulate data

↳ A process performed by VPN service that protect my data by wrapping sensitive data in other data packets

Security Zone

A segment of network that protect the internal framework from the internet



Overview of subnetting

Subnetting is the process of taking one large network and **dividing it into several smaller, organized groups** called **subnets**. Think of it like this: If your company's network is a large city, each subnet is a distinct neighborhood within that city.

Subnetting divides up a network address range into smaller subnets within the network. These organized subnets are defined by the unique combination of the **IP address** and the **network mask** assigned to every device, effectively creating a "network within a network." Subnetting creates a network of devices to function as their own network. This makes the network more efficient and can also be used to create security zones. If devices on the same subnet communicate with each other, the switch changes the transmissions to stay on the same subnet, improving speed and efficiency of the communications.

Proxy server

A server that fulfills the request of a client by forwarding them to other services.

types of Proxy server

Forward proxy server : Regulates and restricts a person's access to the Internet

Reverse proxy server

Regulate & restricts the internet access to internal server

Remote access and site-to-site VPNs

Individual users use remote access VPNs to establish a connection between a personal device and a VPN server. Remote access VPNs encrypt data sent or received through a personal device. The connection between the user and the remote access VPN is established through the internet.

Enterprises use site-to-site VPNs largely to extend their network to other networks and locations. This is particularly useful for organizations that have many offices across the globe. IPSec is commonly used in site-to-site VPNs to create an encrypted tunnel between the primary network and the remote network. One disadvantage of site-to-site VPNs is how complex they can be to configure and manage compared to remote VPNs.

WireGuard VPN vs. IPSec VPN

WireGuard and IPSec are two different VPN protocols used to encrypt traffic over a secure network tunnel. The majority of VPN providers offer a variety of options for VPN protocols, such as WireGuard or IPSec. Ultimately, choosing between IPSec and WireGuard depends on many factors, including connection speeds, compatibility with existing network infrastructure, and business or individual needs.

WireGuard VPN

WireGuard is a high-speed VPN protocol, with advanced encryption, to protect users when they are accessing the internet. It's designed to be simple to set up and maintain. WireGuard can be used for both site-to-site connection and client-server connections. WireGuard is relatively newer than IPSec, and is used by many people due to the fact that its download speed is enhanced by using fewer lines of code. WireGuard is also open source, which makes it easier for users to deploy and debug. This protocol is useful for processes that require faster download speeds, such as streaming video content or downloading large files.

IPSec VPN

IPSec is another VPN protocol that may be used to set up VPNs. Most VPN providers use IPSec to encrypt and authenticate data packets in order to establish secure, encrypted connections. Since IPSec is one of the earlier VPN protocols, many operating systems support IPSec from VPN providers.

Module 3

(DOS) Denial of Service attack

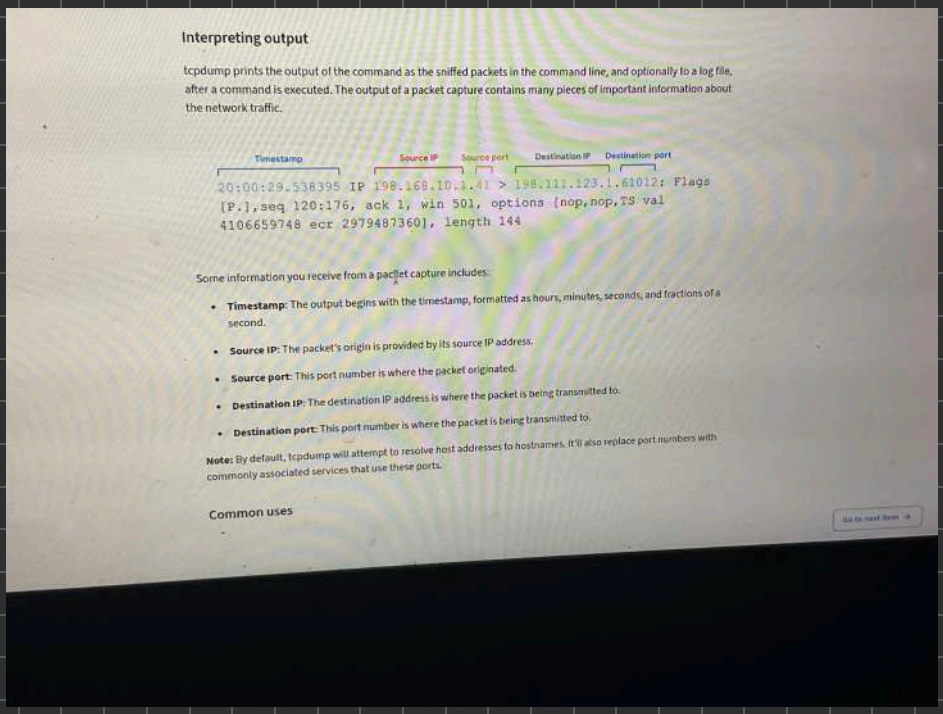
- an attack that floods network with trafic

(DDOS) - use multiple device & location and does same thing

SYN flood attack

- A type of DDOS attack that simulate TCP connection and flood a server with SYN Packets

ICMP - use like status update



Interpreting output

tcpdump prints the output of the command as the sniffed packets in the command line, and optionally to a log file, after a command is executed. The output of a packet capture contains many pieces of important information about the network traffic.

Timestamp | Source IP | Source port | Destination IP | Destination port

```
20:00:29.538395 IP 198.168.10.1.41 > 198.111.123.1.61012: Flags
[P.],seq 120:176, ack 1, win 501, options [nop,nop,TS val
4106659748 ecr 2979487360], length 144
```

Some information you receive from a packet capture includes:

- **Timestamp:** The output begins with the timestamp, formatted as hours, minutes, seconds, and fractions of a second.
- **Source IP:** The packet's origin is provided by its source IP address.
- **Source port:** This port number is where the packet originated.
- **Destination IP:** The destination IP address is where the packet is being transmitted to.
- **Destination port:** This port number is where the packet is being transmitted to.

Note: By default, tcpdump will attempt to resolve host addresses to hostnames. It'll also replace port numbers with commonly associated services that use these ports.

Common uses

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Packet Sniffing Passive

A type of attack where data pack are read in transit

Active Packet Sniffing

A type of attack where data Packets are manipulated in transit.

IP Spoofing

when attacker change the source IP of a data packet to impersonate an unauthorized system and gain access to network

types of common IP spoofing attack

- on-Path attack - Middle to intercept
- Replay attack - delay or repeat packet
- smurf attack - dos + IP

Module -4

security hardening - A process of strengthening a system to reduce its vulnerability and attack surface

Attack Surface

All the potential vulnerabilities that a threat actor could exploit

OS hardening Practice

Patch update : A software and operating system update that address security vulnerability within product or program

Baseline Configuration
:- A documented set of specification within a system that is used as a basis for future builds, releases and update

Network security hardening

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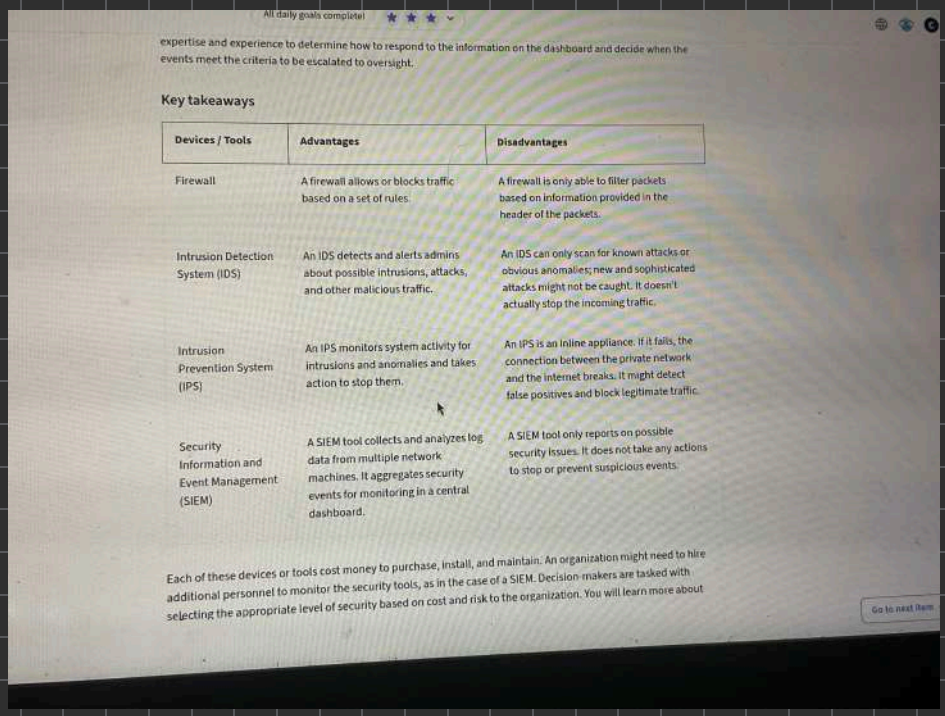

All daily goals complete!

expertise and experience to determine how to respond to the information on the dashboard and decide when the events meet the criteria to be escalated to oversight.

Key takeaways

Devices / Tools	Advantages	Disadvantages
Firewall	A firewall allows or blocks traffic based on a set of rules.	A firewall is only able to filter packets based on information provided in the header of the packets.
Intrusion Detection System (IDS)	An IDS detects and alerts admins about possible intrusions, attacks, and other malicious traffic.	An IDS can only scan for known attacks or obvious anomalies; new and sophisticated attacks might not be caught. It doesn't actually stop the incoming traffic.
Intrusion Prevention System (IPS)	An IPS monitors system activity for intrusions and anomalies and takes action to stop them.	An IPS is an inline appliance. If it fails, the connection between the private network and the internet breaks. It might detect false positives and block legitimate traffic.
Security Information and Event Management (SIEM)	A SIEM tool collects and analyzes log data from multiple network machines. It aggregates security events for monitoring in a central dashboard.	A SIEM tool only reports on possible security issues. It does not take any actions to stop or prevent suspicious events.

Each of these devices or tools cost money to purchase, install, and maintain. An organization might need to hire additional personnel to monitor the security tools, as in the case of a SIEM. Decision-makers are tasked with selecting the appropriate level of security based on cost and risk to the organization. You will learn more about

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